

**BS119**

2026 Fall Semester  
Practice Exercises

**Chapter 13 & Chapter 14**

*AI may assist you, but it can fail you as well!*

□ Use these values where needed

### Molal Boiling-Point-Elevation and Freezing-Point-Depression Constants

| Solvent            | Formula                          | Normal Boiling |              | Normal Freezing |              |
|--------------------|----------------------------------|----------------|--------------|-----------------|--------------|
|                    |                                  | Point (°C)     | $K_b$ (°C/m) | Point (°C)      | $K_f$ (°C/m) |
| Water              | H <sub>2</sub> O                 | 100            | 0.51         | 0.0             | 1.86         |
| Benzene            | C <sub>6</sub> H <sub>6</sub>    | 80.1           | 2.53         | 5.5             | 5.12         |
| Ethanol            | C <sub>2</sub> H <sub>5</sub> OH | 78.4           | 1.22         | -114.6          | 1.99         |
| Tetrachloromethane | CCl <sub>4</sub>                 | 76.8           | 5.02         | -22.3           | 29.8         |
| Trichloromethane   | CHCl <sub>3</sub>                | 61.2           | 3.63         | -63.5           | 4.68         |

### Numerical Values of the Gas Constant $R$ in Various Units

| Units                     | Numerical Value |
|---------------------------|-----------------|
| L·atm/mol·K               | 0.08206         |
| J/mol·K*                  | 8.314           |
| m <sup>3</sup> ·Pa/mol·K* | 8.314           |
| L·torr/mol·K              | 62.36           |
| L·kPa/mol·K               | 8.314           |

\* SI unit

□ Use these values where needed

## Periodic Table of the Elements

| Main Group Representative Elements |  |  |  |  |  |  |  |  |  |  |  | Main Group Representative Elements |  |  |  |  |  |
|------------------------------------|--|--|--|--|--|--|--|--|--|--|--|------------------------------------|--|--|--|--|--|
| 1A <sup>a</sup>                    |  |  |  |  |  |  |  |  |  |  |  | 8A                                 |  |  |  |  |  |
| 1                                  |  |  |  |  |  |  |  |  |  |  |  | 18                                 |  |  |  |  |  |
| 1                                  |  |  |  |  |  |  |  |  |  |  |  | 2                                  |  |  |  |  |  |
| 1.00794                            |  |  |  |  |  |  |  |  |  |  |  | 4.002602                           |  |  |  |  |  |
| 2                                  |  |  |  |  |  |  |  |  |  |  |  | 10                                 |  |  |  |  |  |
| 6.941                              |  |  |  |  |  |  |  |  |  |  |  | 20.1797                            |  |  |  |  |  |
| 11                                 |  |  |  |  |  |  |  |  |  |  |  | 13                                 |  |  |  |  |  |
| 22.989770                          |  |  |  |  |  |  |  |  |  |  |  | 26.981538                          |  |  |  |  |  |
| 19                                 |  |  |  |  |  |  |  |  |  |  |  | 31                                 |  |  |  |  |  |
| 39.0983                            |  |  |  |  |  |  |  |  |  |  |  | 69.723                             |  |  |  |  |  |
| 37                                 |  |  |  |  |  |  |  |  |  |  |  | 49                                 |  |  |  |  |  |
| 85.4678                            |  |  |  |  |  |  |  |  |  |  |  | 114.818                            |  |  |  |  |  |
| 55                                 |  |  |  |  |  |  |  |  |  |  |  | 81                                 |  |  |  |  |  |
| 132.90545                          |  |  |  |  |  |  |  |  |  |  |  | 204.3833                           |  |  |  |  |  |
| 87                                 |  |  |  |  |  |  |  |  |  |  |  | 113                                |  |  |  |  |  |
| [223.02]                           |  |  |  |  |  |  |  |  |  |  |  | [284]                              |  |  |  |  |  |
| 88                                 |  |  |  |  |  |  |  |  |  |  |  | 114                                |  |  |  |  |  |
| [226.03]                           |  |  |  |  |  |  |  |  |  |  |  | [289.2]                            |  |  |  |  |  |
| 103                                |  |  |  |  |  |  |  |  |  |  |  | 115                                |  |  |  |  |  |
| [262.11]                           |  |  |  |  |  |  |  |  |  |  |  | [288]                              |  |  |  |  |  |
| 104                                |  |  |  |  |  |  |  |  |  |  |  | 116                                |  |  |  |  |  |
| [261.11]                           |  |  |  |  |  |  |  |  |  |  |  | [293]                              |  |  |  |  |  |
| 105                                |  |  |  |  |  |  |  |  |  |  |  | 117                                |  |  |  |  |  |
| [262.11]                           |  |  |  |  |  |  |  |  |  |  |  | [294]                              |  |  |  |  |  |
| 106                                |  |  |  |  |  |  |  |  |  |  |  | 118                                |  |  |  |  |  |
| [266.12]                           |  |  |  |  |  |  |  |  |  |  |  | [294]                              |  |  |  |  |  |
| 107                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [264.12]                           |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| 108                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [269.13]                           |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| 109                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [268.14]                           |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| 110                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [281.15]                           |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| 111                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [272.15]                           |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| 112                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [285]                              |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| 113                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [284]                              |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| 114                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [289.2]                            |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| 115                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [288]                              |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| 116                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [293]                              |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| 117                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [294]                              |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| 118                                |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |
| [294]                              |  |  |  |  |  |  |  |  |  |  |  |                                    |  |  |  |  |  |

|                   |                      |                      |                       |                      |                      |                      |                      |                      |                       |                      |                       |                       |                       |                       |
|-------------------|----------------------|----------------------|-----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|-----------------------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Lanthanide series | 57<br>La<br>138.9055 | 58<br>Ce<br>140.116  | 59<br>Pr<br>140.90765 | 60<br>Nd<br>144.24   | 61<br>Pm<br>[145]    | 62<br>Sm<br>150.36   | 63<br>Eu<br>151.964  | 64<br>Gd<br>157.25   | 65<br>Tb<br>158.92534 | 66<br>Dy<br>162.50   | 67<br>Ho<br>164.93032 | 68<br>Er<br>167.259   | 69<br>Tm<br>168.93421 | 70<br>Yb<br>173.04    |
| Actinide series   | 89<br>Ac<br>[227.03] | 90<br>Th<br>232.0381 | 91<br>Pa<br>231.03588 | 92<br>U<br>238.02891 | 93<br>Np<br>[237.05] | 94<br>Pu<br>[244.06] | 95<br>Am<br>[243.06] | 96<br>Cm<br>[247.07] | 97<br>Bk<br>[247.07]  | 98<br>Cf<br>[251.08] | 99<br>Es<br>[252.08]  | 100<br>Fm<br>[257.10] | 101<br>Md<br>[258.10] | 102<br>No<br>[259.10] |

# Part I. Chapter 13 (Properties of Solutions)

1. Define each of the following terms, phrases, laws, rules, etc. Provide at least one example where needed.

- |                           |                             |                          |
|---------------------------|-----------------------------|--------------------------|
| ✓ Solute                  | ✓ Hydrogen bonding          | ✓ Dissociation           |
| ✓ Solvent                 | ✓ Henry's law               | ✓ Osmosis                |
| ✓ Solution                | ✓ Mass percentage           | ✓ Semipermeable membrane |
| ✓ Solvation               | ✓ ppm                       | ✓ Osmotic pressure       |
| ✓ Hydration               | ✓ ppb                       | ✓ Colloid                |
| ✓ Enthalpy                | ✓ Mole fraction             | ✓ Colloidal dispersion   |
| ✓ Entropy                 | ✓ Molarity                  | ✓ Tyndall effect         |
| ✓ Exothermic              | ✓ Molality                  | ✓ Hydrophilic            |
| ✓ Endothermic             | ✓ Density                   | ✓ Hydrophobic            |
| ✓ Saturated solution      | ✓ Colligative property      | ✓ Adsorption             |
| ✓ Unsaturated solution    | ✓ Raoult's law              | ✓ Brownian motion        |
| ✓ Supersaturated solution | ✓ Ideal solution            |                          |
| ✓ Dynamic equilibrium     | ✓ Vapor-pressure lowering   |                          |
| ✓ Dissolution             | ✓ Boiling-point elevation   |                          |
| ✓ Crystallization         | ✓ Freezing-point depression |                          |
| ✓ Solubility              | ✓ $K_b$                     |                          |
| ✓ Intermolecular forces   | ✓ $K_f$                     |                          |
| ✓ "Like dissolves like"   | ✓ van't Hoff factor ( $i$ ) |                          |
| ✓ Miscible                | ✓ Electrolyte               |                          |
| ✓ Immiscible              | ✓ Nonelectrolyte            |                          |

2. The figure shows two identical volumetric flasks containing the same solution at two temperatures.

- a. Does the molarity of the solution change with the change in temperature?
- b. Does the molality of the solution change with the change in temperature?



3. If you compare the solubilities of the noble gases in water, you find that solubility increases from smallest atomic weight to largest,  $\text{Ar} < \text{Kr} < \text{Xe}$ . Explain.
4. Suppose you take a solution and add more solvent, so that the original mass of solvent is doubled. You take this new solution and add more solute, so that the original mass of the solute is doubled. What happens to the molality of the final solution, compared to the original molality?
- It is doubled.
  - It is decreased by half.
  - It is unchanged.
  - It will increase or decrease depending on the molar mass of the solute.
  - There is no way to tell without knowing the molar mass of the solute.

5. Predict whether each of the following substances is more likely to dissolve in the nonpolar solvent carbon tetrachloride ( $\text{CCl}_4$ ) or in water:
- a.  $\text{C}_7\text{H}_{16}$
  - b.  $\text{Na}_2\text{SO}_4$
  - c.  $\text{HCl}$
  - d.  $\text{I}_2$
6. Indicate the type of solute–solvent interaction that should be most important in each of the following solutions:
- a.  $\text{CCl}_4$  in benzene ( $\text{C}_6\text{H}_6$ )
  - b. Methanol ( $\text{CH}_3\text{OH}$ ) in water
  - c.  $\text{KBr}$  in water
  - d.  $\text{HCl}$  in acetonitrile ( $\text{CH}_3\text{CN}$ ).

7. Which of the following in each pair is likely to be more soluble in hexane,  $\text{C}_6\text{H}_{14}$  (Explain your answer in each case):
- $\text{CCl}_4$  or  $\text{CaCl}_2$
  - Benzene ( $\text{C}_6\text{H}_6$ ) or glycerol,  $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$
  - Octanoic acid,  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$ , or acetic acid,  $\text{CH}_3\text{COOH}$ ?
8. You double the partial pressure of a gas over a liquid at constant temperature. Which of these statements is then true?
- The Henry's law constant is doubled.
  - The Henry's law constant is decreased by half.
  - There are half as many gas molecules in the liquid.
  - There are twice as many gas molecules in the liquid.
  - There is no change in the number of gas molecules in the liquid.

- 9.** How would each of the colligative properties of solutions change if a solute is volatile and dissociating?
- 10.** A solution is made by dissolving 13.5 g of glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ ) in 0.100 kg of water. What is the mass percentage of solute in this solution?
- 11.** A 2.5 g sample of groundwater was found to contain 5.4  $\mu\text{g}$  of  $\text{Zn}^{2+}$ . What is the concentration of  $\text{Zn}^{2+}$  in parts per million?

**12.** A commercial bleaching solution contains 3.62% by mass of sodium hypochlorite, NaOCl. What is the mass of NaOCl in a bottle containing 2.50 kg of bleaching solution?

- 13.** Which of the following actions will raise the osmotic pressure of a solution?
- a. Decreasing the solute concentration
  - b. Decreasing the temperature
  - c. Adding more solvent
  - d. Increasing the temperature
  - e. None of the above

**14.** An aqueous solution of hydrochloric acid contains 36% HCl by mass.

- a. Calculate the mole fraction of HCl in the solution.
- b. Calculate the molality of HCl in the solution.

**15.** Some soft drinks contain up to 85 ppm oxygen.

- a. What is this concentration in mol/L?
- b. What partial pressure of O<sub>2</sub> above water is needed to obtain 85 ppm O<sub>2</sub> in water at 10 °C? (The Henry's law constant for O<sub>2</sub> at this temperature is  $1.69 \times 10^{-5} \text{ mol/m}^3 \text{ Pa}$ .)

- 16.** Calculate the concentration of  $\text{CO}_2$  in a soft drink that is bottled with a partial pressure of  $\text{CO}_2$  of 4.0 atm over the liquid at 25 °C. The Henry's law constant for  $\text{CO}_2$  in water at this temperature is  $3.4 \times 10^{-2}$  mol/L-atm.
- 17.** A solution is made containing 50.0 g of ethanol ( $\text{C}_2\text{H}_5\text{OH}$ ) in 1000 g of  $\text{H}_2\text{O}$ . Calculate:
- The mole fraction of  $\text{C}_2\text{H}_5\text{OH}$
  - The mass percent of  $\text{C}_2\text{H}_5\text{OH}$
  - The molality of  $\text{C}_2\text{H}_5\text{OH}$

- 18.** Acetonitrile ( $\text{CH}_3\text{CN}$ ) is a polar organic solvent that dissolves a wide range of solutes, including many salts. The density of a 1.80 M LiBr solution in acetonitrile is  $0.826 \text{ g/cm}^3$ . Calculate the concentration of the solution in:
- Molality
  - Mole fraction of LiBr
  - Mass percentage of  $\text{CH}_3\text{CN}$ .
- 19.** A sulfuric acid solution containing 697.6 g of  $\text{H}_2\text{SO}_4$  per liter of solution has a density of  $1.395 \text{ g/cm}^3$ . Calculate
- The mass percentage.
  - The mole fraction.
  - The molality
  - The molarity of  $\text{H}_2\text{SO}_4$  in this solution.

- 20.** The density of acetonitrile ( $\text{CH}_3\text{CN}$ ) is 0.786 g/mL and the density of methanol ( $\text{CH}_3\text{OH}$ ) is 0.791 g/mL. A solution is made by dissolving 25.0 mL of  $\text{CH}_3\text{OH}$  in 100 mL of  $\text{CH}_3\text{CN}$ .
- What is the mole fraction of methanol in the solution?
  - What is the molality of the solution?
  - Assuming that the volumes are additive, what is the molarity of  $\text{CH}_3\text{OH}$  in the solution?
- 21.** Describe how you would prepare each of the following aqueous solutions:
- 1.50 L of 0.110 M  $(\text{NH}_4)_2\text{SO}_4$  solution, starting with solid  $(\text{NH}_4)_2\text{SO}_4$ .
  - 225 g of a solution that is 0.65 m in  $\text{Na}_2\text{CO}_3$ , starting with the solid solute.
  - 1.20 L of a solution that is 15.0%  $\text{Pb}(\text{NO}_3)_2$  by mass (the density of the solution is 1.16 g/mL), starting with solid solute.
  - A 0.50 M solution of  $\text{HCl}$  that would just neutralize 5.5 g of  $\text{Ba}(\text{OH})_2$  starting with 6.0 M  $\text{HCl}$ .

- 22.** Consider two solutions, one formed by adding 150 g of glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ ) to 1 L of water and the other formed by adding 150 g of sucrose ( $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ ) to 1 L of water. Calculate the vapor pressure for each solution at 25 °C. The vapor pressure of pure water at this temperature is 3.17 kPa.
- 23.** Lauryl alcohol is obtained from coconut oil and is used to make detergents. A solution of 5.00 g of lauryl alcohol in 0.100 kg of benzene freezes at 4.1 °C. What is the molar mass of lauryl alcohol from this data?
- 24.** The osmotic pressure of a 0.010 M aqueous solution of  $\text{CaCl}_2$  is found to be 68.3 kPa at 25 °C. Calculate the van't Hoff factor,  $i$ , for the solution.

- 25.** The normal boiling point of ethanol,  $\text{CH}_3\text{CH}_2\text{OH}$ , is  $78.4\text{ }^\circ\text{C}$ . When  $3.26\text{ g}$  of a soluble nonelectrolyte is dissolved in  $100.0\text{ g}$  of ethanol at that temperature, the vapor pressure of the solution is  $100\text{ kPa}$ . What is the molar mass of the solute?
- 26.** Benzene  $\text{C}_6\text{H}_6$  boils at  $80.1\text{ }^\circ\text{C}$  and has a density of  $0.876\text{ g/mL}$ .
- When  $0.100\text{ mol}$  of a non-dissociating solute is dissolved in  $500\text{ mL}$  of  $\text{C}_6\text{H}_6$ , the solution boils at  $79.52\text{ }^\circ\text{C}$ . What is the molal boiling-point-elevation constant for  $\text{C}_6\text{H}_6$ ?
  - When  $10.0\text{ g}$  of a non-dissociating unknown is dissolved in  $500\text{ mL}$  of  $\text{C}_6\text{H}_6$ , the solution boils at  $79.23\text{ }^\circ\text{C}$ . What is the molar mass of the unknown?
- 27.** What is the osmotic pressure, in atm, of a  $0.0020\text{ M}$  sucrose ( $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ ) solution at  $20.0\text{ }^\circ\text{C}$  ?

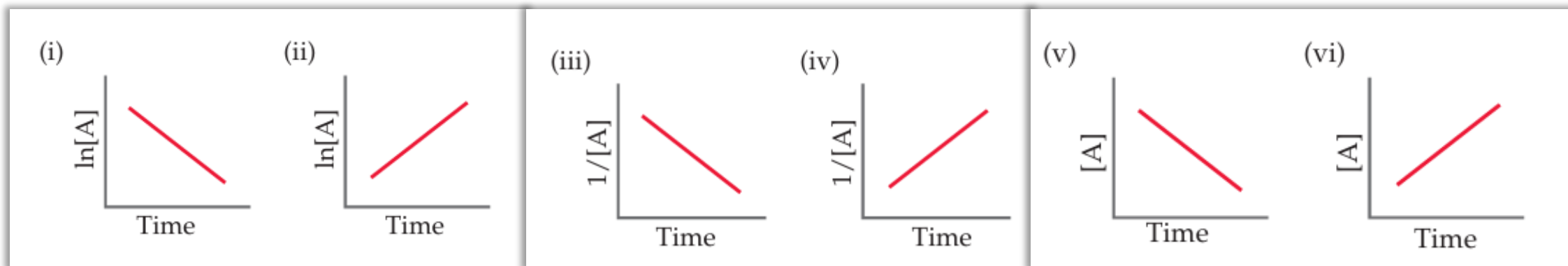
- 28.** The phrase "like dissolves like" refers to the fact that \_\_\_\_\_.  
A. gases can only dissolve other gases  
B. polar solvents dissolve polar solutes and nonpolar solvents dissolve nonpolar solutes  
C. solvents can only dissolve solutes of similar molar mass  
D. condensed phases can only dissolve other condensed phases  
E. polar solvents dissolve nonpolar solutes and vice versa
- 29.** Which of the following choices has the compounds correctly arranged in order of increasing solubility in water? (least soluble to most soluble)  
A.  $\text{CCl}_4 < \text{CHCl}_3 < \text{NaNO}_3$   
B.  $\text{CH}_3\text{OH} < \text{CH}_4 < \text{LiF}$   
C.  $\text{CH}_4 < \text{NaNO}_3 < \text{CHCl}_3$   
D.  $\text{LiF} < \text{NaNO}_3 < \text{CHCl}_3$   
E.  $\text{CH}_3\text{OH} < \text{Cl}_4 < \text{CHCl}_3$
- 30.** A solution is prepared by dissolving 15.0 g of  $\text{NH}_3$  in 250.0 g of water. The density of the resulting solution is 0.974 g/mL. The molarity of  $\text{NH}_3$  in the solution is \_\_\_\_\_ M.  
A. 0.00353                      C. 60.0                      E. 3.53  
B. 0.882                         D. 3.24

# Part II. Chapter 14 (Chemical Kinetics)

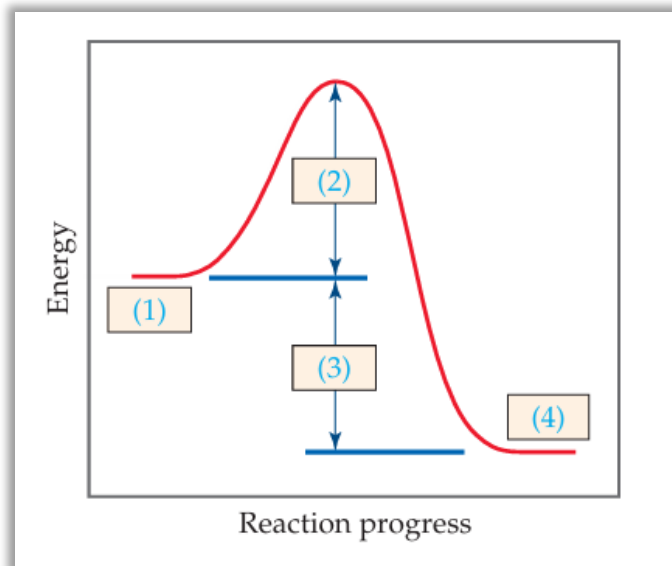
1. Define each of the following terms, phrases, laws, rules, etc. Provide at least one example where needed.

- |                          |                            |                          |
|--------------------------|----------------------------|--------------------------|
| ✓ Chemical kinetics      | ✓ Kinetic energy           | ✓ Homogeneous catalyst   |
| ✓ Reaction rate          | ✓ Activation energy        | ✓ Heterogeneous catalyst |
| ✓ Average rate           | ✓ Activated complex        | ✓ Adsorption             |
| ✓ Instantaneous rate     | ✓ Transition state         | ✓ Catalyst surface       |
| ✓ Rate of disappearance  | ✓ Effective collision      | ✓ Enzyme                 |
| ✓ Rate of appearance     | ✓ Molecular orientation    | ✓ Substrate              |
| ✓ Stoichiometry          | ✓ Arrhenius equation       | ✓ Active site            |
| ✓ Rate law               | ✓ Frequency factor ( $A$ ) | ✓ Lock-and-key model     |
| ✓ Rate constant ( $k$ )  | ✓ Reaction mechanism       |                          |
| ✓ Reaction order         | ✓ Elementary reaction      |                          |
| ✓ Overall reaction order | ✓ Molecularity             |                          |
| ✓ Zero-order reaction    | ✓ Unimolecular             |                          |
| ✓ First-order reaction   | ✓ Bimolecular              |                          |
| ✓ Second-order reaction  | ✓ Termolecular             |                          |
| ✓ Integrated rate law    | ✓ Intermediate             |                          |
| ✓ Half-life              | ✓ Rate-determining step    |                          |
| ✓ Spectroscopy           | ✓ Multistep mechanism      |                          |
| ✓ Beer's law             | ✓ Catalyst                 |                          |
| ✓ Collision model        | ✓ Catalysis                |                          |

2. Which of the following linear plots do you expect for a reaction  $A \rightarrow \text{products}$  if the kinetics are (a) zero order, (b) first order, or (c) second order?

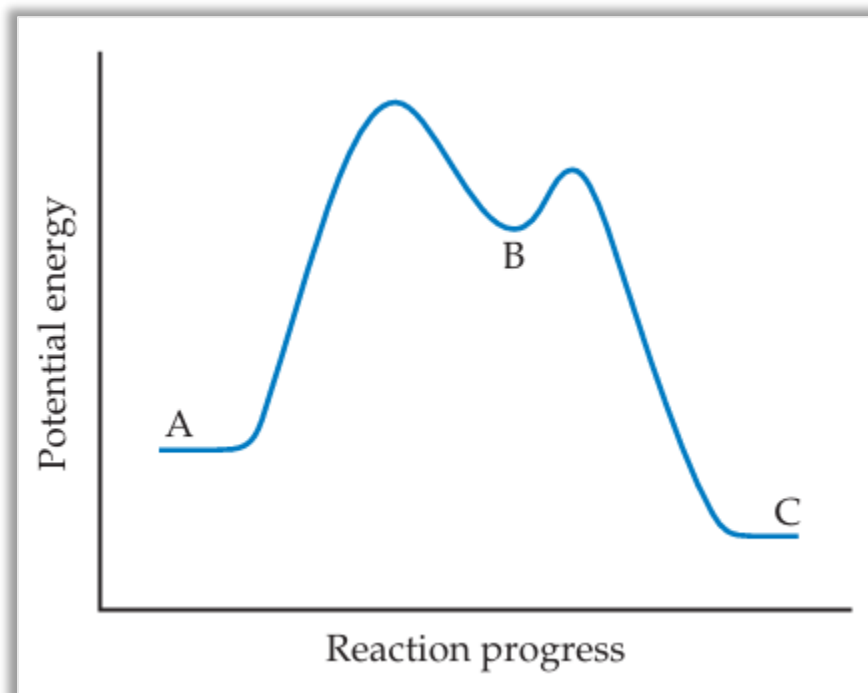


3. The following diagram shows a reaction profile. Label the components indicated by the boxes



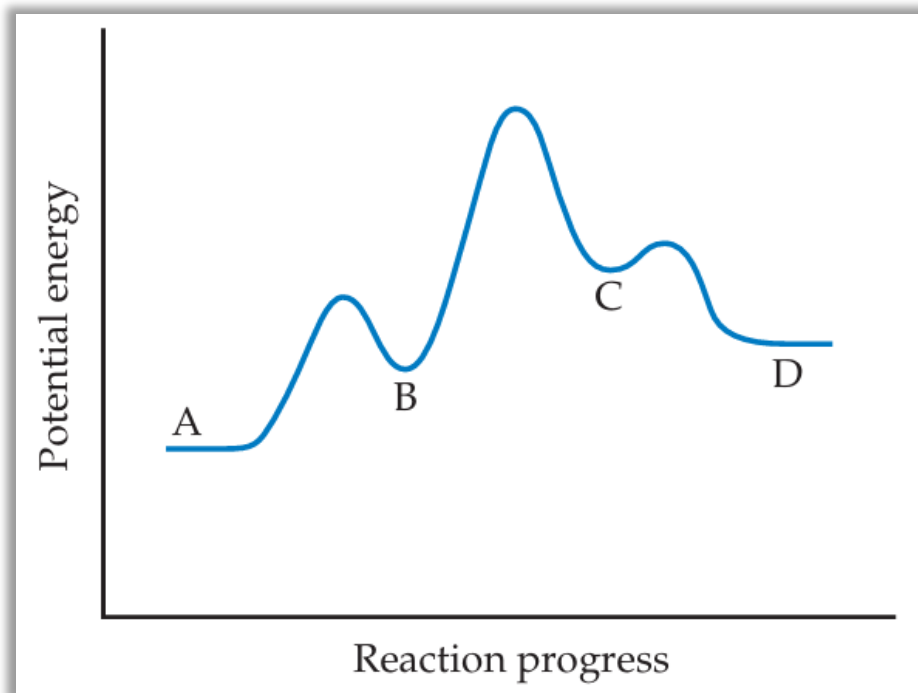
4. Based on the reaction profile:

- How many intermediates are formed in the reaction  $A \rightarrow C$ ?
- How many transition states are there?
- Which step,  $A \rightarrow B$  or  $B \rightarrow C$ , is the faster?
- For the reaction  $A \rightarrow C$ , is  $\Delta E$  positive, negative, or zero?



5. Based on the following reaction profile:

- How many intermediates are formed in the reaction  $A \rightarrow D$ ?
- How many transition states are there?
- Which step is the fastest?
- For the reaction  $A \rightarrow D$ , is  $\Delta E$  positive, negative, or zero?



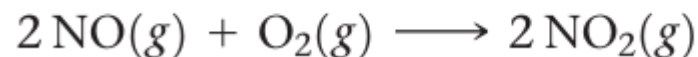
6. Based on their activation energies and energy changes and assuming that all collision factors are the same, rank the following reactions from slowest to fastest.

(a)  $E_a = 75 \text{ kJ/mol}$ ;  $\Delta E = -20 \text{ kJ/mol}$

(b)  $E_a = 100 \text{ kJ/mol}$ ;  $\Delta E = +30 \text{ kJ/mol}$

(c)  $E_a = 85 \text{ kJ/mol}$ ;  $\Delta E = -50 \text{ kJ/mol}$

7. The oxidation of  $\text{SO}_2$  to  $\text{SO}_3$  is accelerated by  $\text{NO}_2$ . The reaction proceeds according to:



- Show that, with appropriate coefficients, the two reactions can be summed to give the overall oxidation of  $\text{SO}_2$  by  $\text{O}_2$  to give  $\text{SO}_3$ .
- Do we consider  $\text{NO}_2$  a catalyst or an intermediate in this reaction?
- Would you classify  $\text{NO}$  as a catalyst or as an intermediate?
- Is this an example of homogeneous catalysis or heterogeneous catalysis?

8. The isomerization of methyl isonitrile ( $\text{CH}_3\text{NC}$ ) to acetonitrile ( $\text{CH}_3\text{CN}$ ) was studied in the gas phase at  $215^\circ\text{C}$ , and the following data were obtained:

| Time (s) | $[\text{CH}_3\text{NC}]$ (M) |
|----------|------------------------------|
| 0        | 0.0165                       |
| 2000     | 0.0110                       |
| 5000     | 0.00591                      |
| 8000     | 0.00314                      |
| 12,000   | 0.00137                      |
| 15,000   | 0.00074                      |

- Calculate the average rate of reaction, in  $M/s$ , for the time interval between each measurement.
- Calculate the average rate of reaction over the entire time of the data from  $t = 0$  to  $t = 15,000$  s.
- Which is greater, the average rate between  $t = 2000$  and  $t = 12,000$  s, or between  $t = 8000$  and  $t = 15,000$  s?
- Graph  $[\text{CH}_3\text{NC}]$  versus time and determine the instantaneous rates in  $M/s$  at  $t = 5000$  s and  $t = 8000$  s.

9. Consider the following reaction:



The rate law for this reaction is first order in  $\text{C}_2\text{H}_5\text{Cl}$  and first order in  $\text{OH}^-$ . When  $[\text{C}_2\text{H}_5\text{Cl}] = 4.0 \times 10^{-3} \text{ M}$  and  $[\text{OH}^-] = 2.5 \times 10^{-2} \text{ M}$ , the reaction rate at 310 K is  $5.20 \times 10^{-2} \text{ M/s}$ .

- What is the value of the rate constant?
- What are the units of the rate constant?
- What would happen to the rate if the concentration of  $\text{OH}^-$  were cut in half?
- What would happen to the rate if the concentration of both reactants were cut in half?

10. The following data were measured for the reaction  $\text{BF}_3(\text{g}) + \text{NH}_3(\text{g}) \rightarrow \text{F}_3\text{BNH}_3(\text{g})$ .

| Experiment | $[\text{BF}_3] \text{ (M)}$ | $[\text{NH}_3] \text{ (M)}$ | Initial Rate (M/s) |
|------------|-----------------------------|-----------------------------|--------------------|
| 1          | 0.250                       | 0.250                       | 0.2130             |
| 2          | 0.250                       | 0.125                       | 0.1065             |
| 3          | 0.200                       | 0.100                       | 0.0682             |
| 4          | 0.350                       | 0.100                       | 0.1193             |
| 5          | 0.175                       | 0.100                       | 0.0596             |

- What is the rate law for the reaction?
- What is the overall order of the reaction?
- Calculate the rate constant with proper units.
- What is the rate when  $[\text{BF}_3] = 0.100 \text{ M}$  and  $[\text{NH}_3] = 0.500 \text{ M}$ ?

**11.** Consider the gas-phase reaction between nitric oxide and bromine at 273 °C:

**$2\text{NO}(\text{g}) + \text{Br}_2(\text{g}) \rightarrow 2\text{NOBr}(\text{g})$** . The following data for the initial rate of appearance of NOBr were obtained:

| Experiment | [NO] (M) | [Br <sub>2</sub> ] (M) | Initial Rate (M/s) |
|------------|----------|------------------------|--------------------|
| 1          | 0.10     | 0.20                   | 24                 |
| 2          | 0.25     | 0.20                   | 150                |
| 3          | 0.10     | 0.50                   | 60                 |
| 4          | 0.35     | 0.50                   | 735                |

- Determine the rate law.
- Calculate the average value of the rate constant for the appearance of NOBr from the four data sets.
- How is the rate of appearance of NOBr related to the rate of disappearance of Br<sub>2</sub>?
- What is the rate of disappearance of Br<sub>2</sub> when [NO] = 0.075 M and [Br<sub>2</sub>] = 0.25 M?

- 12.** Consider the reaction  $2\text{C}_4\text{H}_6(\text{g}) \rightarrow \text{C}_8\text{H}_{12}(\text{g})$ , the rate constant at 500 K is  $1.4 \times 10^{-2} \text{ M}^{-1} \text{ s}^{-1}$  and the initial concentration of A is 0.020 M.
- What is the order of the reaction?
  - What is the half-life for this reaction?
  - How much  $\text{C}_4\text{H}_6(\text{g})$  will remain after 1 hour at the same temperature?
- 13.** The gas-phase decomposition of  $\text{NO}_2$ ,  $2\text{NO}_2(\text{g}) \rightarrow 2\text{NO}(\text{g}) + \text{O}_2(\text{g})$ , is studied at 383 °C, giving the following data:

| Time (s) | $[\text{NO}_2] \text{ (M)}$ |
|----------|-----------------------------|
| 0.0      | 0.100                       |
| 5.0      | 0.017                       |
| 10.0     | 0.0090                      |
| 15.0     | 0.0062                      |
| 20.0     | 0.0047                      |

- Is the reaction first-order or second-order with respect to the concentration of  $\text{NO}_2$ ?
- What is the rate constant?
- Predict the reaction rates at the beginning of the reaction for initial concentrations of 0.200 M, 0.100 M, and 0.050 M  $\text{NO}_2$ .

14. Consider the following hypothetical aqueous reaction:  **$A(aq) \rightarrow B(aq)$** . A flask is charged with 0.065 mol of A in a total volume of 100.0 mL. The following data are collected.

| Time (min) | 0     | 10    | 20    | 30    | 40    |
|------------|-------|-------|-------|-------|-------|
| Moles of A | 0.065 | 0.051 | 0.042 | 0.036 | 0.031 |

- Calculate the number of moles of B at each time in the table, assuming that there are no molecules of B at time zero and that A cleanly converts to B with no intermediates.
- Calculate the average rate of disappearance of A for each 10 min interval in units of  $M/s$ .
- Between  $t = 0$  min and  $t = 30$  min, what is the average rate of appearance of B in units of  $M/s$ ? Assume that the volume of the solution is constant.
- Using appropriate graphs, determine the reaction order with respect to A.
- How long will it take to consume 90% of A initially present in the flask?

- 15.** The rate of the following reaction was measured at several temperatures as shown in the table. Calculate the value of  $E_a$  by constructing an appropriate graph.

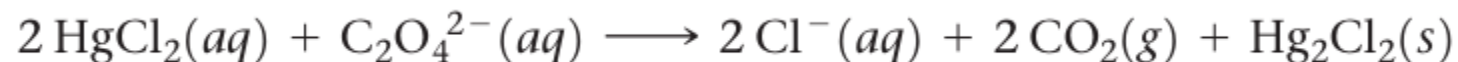


| Temperature ( $^{\circ}\text{C}$ ) | $k (M^{-1} s^{-1})$ |
|------------------------------------|---------------------|
| 15                                 | 0.0521              |
| 25                                 | 0.101               |
| 35                                 | 0.184               |
| 45                                 | 0.332               |

- 16.** The first-order rate constant for reaction of a particular organic compound with water varies with temperature as shown in the following table. From these data, calculate the activation energy in units of  $\text{kJ/mol}$ .

| Temperature (K) | Rate Constant ( $s^{-1}$ ) |
|-----------------|----------------------------|
| 300             | $3.2 \times 10^{-11}$      |
| 320             | $1.0 \times 10^{-9}$       |
| 340             | $3.0 \times 10^{-8}$       |
| 355             | $2.4 \times 10^{-7}$       |

**17.** Consider the following reaction between mercury (II) chloride and oxalate ion:

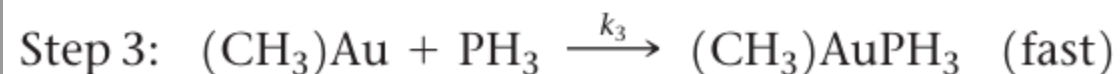
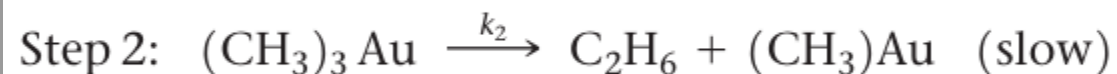
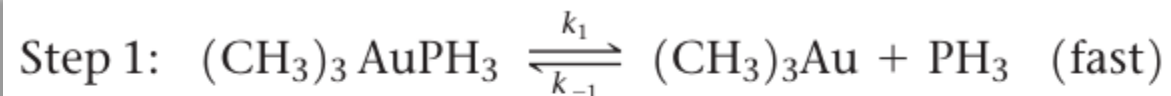


The initial rate of this reaction was determined for several concentrations of  $\text{HgCl}_2$  and  $\text{C}_2\text{O}_4^{2-}$ , and the following rate data were obtained for the rate of disappearance of  $\text{C}_2\text{O}_4^{2-}$ :

| Experiment | $[\text{HgCl}_2] (M)$ | $[\text{C}_2\text{O}_4^{2-}] (M)$ | Rate ( $M/s$ )       |
|------------|-----------------------|-----------------------------------|----------------------|
| 1          | 0.164                 | 0.15                              | $3.2 \times 10^{-5}$ |
| 2          | 0.164                 | 0.45                              | $2.9 \times 10^{-4}$ |
| 3          | 0.082                 | 0.45                              | $1.4 \times 10^{-4}$ |
| 4          | 0.246                 | 0.15                              | $4.8 \times 10^{-5}$ |

- What is the rate law for this reaction?
- What is the value of the rate constant with proper units?
- What is the reaction rate when the initial concentration of  $\text{HgCl}_2$  is  $0.100 M$  and that of  $\text{C}_2\text{O}_4^{2-}$  is  $0.25 M$  if the temperature is the same as that used to obtain the data shown?

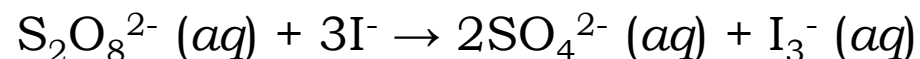
- 18.** In a hydrocarbon solution, the gold compound  $(\text{CH}_3)_3\text{AuPH}_3$  decomposes into ethane ( $\text{C}_2\text{H}_6$ ) and a different gold compound,  $(\text{CH}_3)\text{AuPH}_3$ . The following mechanism has been proposed for the decomposition of  $(\text{CH}_3)_3\text{AuPH}_3$



- What is the overall reaction?
- What are the intermediates in the mechanism?
- What is the molecularity of each of the elementary steps?
- What is the rate-determining step?
- What is the rate law predicted by this mechanism?
- What would be the effect on the reaction rate of adding  $\text{PH}_3$  to the solution of  $(\text{CH}_3)_3\text{AuPH}_3$ ?

- 19.** The reaction between ethyl iodide and hydroxide ion in ethanol ( $\text{C}_2\text{H}_5\text{OH}$ ) solution,  $\text{C}_2\text{H}_5\text{I}(\text{alc}) + \text{OH}^{-1}(\text{alc}) \rightarrow \text{C}_2\text{H}_5\text{OH}(\text{l}) + \text{I}^{-1}(\text{alc})$  has an activation energy of  $86.8 \text{ kJ/mol}$  and a frequency factor of  $2.1 \times 10^{11} \text{ M}^{-1}\text{s}^{-1}$ .
- Predict the rate constant for the reaction at  $30^\circ\text{C}$ .
  - A solution of KOH in ethanol is made up by dissolving 0.500 g KOH in ethanol to form 500 mL of solution. Similarly, 1.500 g of  $\text{C}_2\text{H}_5\text{I}$  is dissolved in ethanol to form 500 mL of solution. Equal volumes of the two solutions are mixed. Assuming the reaction is first order in each reactant, what is the initial rate at  $30^\circ\text{C}$ ?
  - Which reagent in the reaction is limiting, assuming the reaction proceeds to completion?
  - Assuming the frequency factor and activation energy do not change as a function of temperature, calculate the rate constant for the reaction at  $40^\circ\text{C}$ .

- 20.** The peroxydisulfate ion ( $\text{S}_2\text{O}_8^{2-}$ ) reacts with the iodide ion in aqueous solution via the reaction:



An aqueous solution containing  $0.050\text{ M}$  of  $\text{S}_2\text{O}_8^{2-}$  ion and  $0.072\text{ M}$  of  $\text{I}^-$  is prepared, and the progress of the reaction followed by measuring  $[\text{I}^-]$ . The data obtained is given in the following table.

|                    |       |       |       |        |        |
|--------------------|-------|-------|-------|--------|--------|
| Time (s)           | 0.000 | 400.0 | 800.0 | 1200.0 | 1600.0 |
| $[\text{I}^-]$ (M) | 0.072 | 0.057 | 0.046 | 0.037  | 0.029  |

- The average rate of disappearance of  $\text{I}^-$  between  $400.0\text{ s}$  and  $800.0\text{ s}$  is \_\_\_\_\_  $\text{M/s}$ .
- What is the average rate of disappearance ( $\text{M/s}$ ) of  $\text{I}^-$  in the initial  $800.0\text{ s}$ ?
- What is the average rate of disappearance ( $\text{M/s}$ ) of  $\text{I}^-$  between  $800.0\text{ s}$  and  $1200.0\text{ s}$ ?
- What is the concentration ( $\text{M}$ ) of  $\text{S}_2\text{O}_8^{2-}$  remaining at  $1200\text{ s}$ ?
- The concentration of  $\text{S}_2\text{O}_8^{2-}$  remaining at  $800\text{ s}$  is \_\_\_\_\_  $\text{M}$ .
- What is the concentration ( $\text{M}$ ) of  $\text{S}_2\text{O}_8^{2-}$  remaining at  $t = 0$ ?

- 21.** What is the half-life of an unknown compound if 17.0% of the compound decomposes in 150 minutes via a first-order process?
- A. 59 minutes                      C. 116 minutes                      E. 558 minutes  
B. 40 minutes                      D. 279 minutes
- 22.** The rate constant for a particular second-order reaction is  $0.47 \text{ M}^{-1}\text{s}^{-1}$ . If the initial concentration of reactant  $0.25 \text{ mol/L}$ , it takes \_\_\_\_\_ s for the concentration to decrease to  $0.20 \text{ mol/L}$ .
- A. 2.1              B. 1.4              C. 1.0              D. 0.47              E. 0.20
- 23.** What is the magnitude of  $k$  for a first-order reaction at  $75.0^\circ\text{C}$  if  $E_a = 95.1 \text{ kJ/mol}$  and rate constant is  $1.35 \times 10^2 \text{ s}^{-1}$  at  $25.0^\circ\text{C}$ ?
- A.  $3.36 \times 10^4$               B.  $3.85 \times 10^4$               C. 745              D.  $4.28 \times 10^6$               E.  $1.36 \times 10^2$
- 24.** A compound decomposes by a first-order process. If 13% of the compound decomposes in 60 minutes, the half-life of the compound is \_\_\_\_\_ min.
- 25.** A particular first-order reaction has a rate constant of  $1.35 \times 10^2 \text{ s}^{-1}$  at  $25.0^\circ\text{C}$ . What is the magnitude of  $k$  at  $65.0^\circ\text{C}$  if  $E_a = 55.5 \text{ kJ/mol}$ ?

- 26.** Identify all the incorrect statements from the following.
- A. Rates of reaction can be positive or negative.
  - B. The overall reaction order is the sum of the orders of each reactant in the rate law.
  - C. Units of the rate constant of a reaction are independent of the overall reaction order.
  - D. The half-life for a first order rate law depends on the starting concentration.
  - E. The rate limiting step in a reaction is the slowest step in the reaction sequence.
  - F. Heterogeneous catalysts have different phases from reactants.
  - G. The concentration of reactants or products at any time during the reaction can be calculated from the integrated rate law.
  - H. The instantaneous rate of a reaction can be read directly from the graph of molarity versus time at any point on the graph.
  - I. The rate of a second order reaction can depend on the concentrations of more than one reactant.

- 27.** The reaction below is first order in  $[\text{H}_2\text{O}_2]$ :-  $2\text{H}_2\text{O}_2 (\text{l}) \rightarrow 2\text{H}_2\text{O} (\text{l}) + \text{O}_2 (\text{g})$ . A solution originally at  $0.600 \text{ M}$   $\text{H}_2\text{O}_2$  is found to be  $0.075 \text{ M}$  after 54 min. The half-life for this reaction is \_\_\_\_\_ min.
- 28.** The reaction:  $2\text{NOBr} (\text{g}) \rightarrow 2 \text{NO} (\text{g}) + \text{Br}_2 (\text{g})$  is a second-order reaction with a rate constant of  $0.80 \text{ M}^{-1}\text{s}^{-1}$  at  $11^\circ\text{C}$ . If the initial concentration of  $\text{NOBr}$  is  $0.0440 \text{ M}$ , the concentration of  $\text{NOBr}$  after 6.0 seconds is \_\_\_\_\_.
- 29.** The data in the table below were obtained for the reaction:  $\text{A} + \text{B} \rightarrow \text{C}$

| Experiment Number | [A] (M) | [B] (M) | Initial Rate (M/s) |
|-------------------|---------|---------|--------------------|
| 1                 | 0.451   | 0.885   | 1.13               |
| 2                 | 0.451   | 1.77    | 1.13               |
| 3                 | 1.35    | 0.885   | 10.17              |

- a. The rate law for this reaction is rate = \_\_\_\_\_.
- b. The magnitude of the rate constant is \_\_\_\_\_.

- 30.** The concentration of compound A was recorded in the following graph as it decomposed to form B. Predict whether the decomposition is a zero-, first- or second-order reaction. Explain your answer.

